

Top 5 Failures

Failure analysis preserved from the supplied Markdown document.

5. Top 5 Failures

The following failures were selected from the 25 incorrect predictions in the 50-case held-out test set.

Failure 1 — Out-of-scope message classified as delivery status

Case 18: The customer message concerns an address-related issue, but the classifier predicted `delivery_status` instead of `out_of_scope`.

Hypothesis: The classifier appears to have focused on delivery-related vocabulary without correctly recognizing that the request was outside the supported scope.

Failure 2 — Delivery problem classified as refund pending

Case 55: The message contains both delivery and refund-related signals, including the absence of delivery and refund.

Hypothesis: Multiple issue signals created an ambiguous boundary between `delivery_delay_or_missing` and `refund_pending`.

Failure 3 — Return request classified as delivery status

Case 59: The message contains language about an item not being dispatched yet, while the labelled intent is `return_request`.

Hypothesis: Delivery-status vocabulary dominated the classifier even though the intended customer-support action was related to a return.

Failure 4 — Wrong/damaged item classified as delivery delay

Case 60: The customer reports a missing item within a multi-order delivery.

Hypothesis: The classifier interpreted the missing-item wording as a delivery problem instead of recognizing it as a `wrong_or_damaged_item` case.

Failure 5 — Delivery delay classified as return request

Case 72: The message describes an item as lost/damaged and returned to the vendor.

Hypothesis: The word "returned" caused confusion between `delivery_delay_or_missing` and `return_request`.

Common Failure Pattern

The main weakness is **boundary classification** rather than completely unrelated predictions. Several errors occur between closely related support intents, particularly delivery status vs. delivery delay, delivery delay vs. return, and

delivery delay vs. refund. Out-of-scope messages containing support-related vocabulary also cause false in-scope predictions.